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Prevalence and correlates of mixed-handedness in schizophrenia

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Abstract

Increased rates of nonright-handedness have been reported in schizophrenia, but a clear distinction has not been made between left- and mixed-handedness. Handedness preferences in 96 patients fulfilling *DSM-III* criteria for schizophrenia and 43 normal comparison subjects were assessed with the Edinburgh Handedness Inventory. A 100% criterion was used to establish left- and right-handedness. Results were analyzed with Mantel-Haenszel odds ratios adjusted for age and sex. The schizophrenic group showed a significant increase in the proportion of mixed-handers compared with the normal group. There was no increase in pure left-handedness in the schizophrenic relative to the normal group. Mixed-handedness in the schizophrenic patients was significantly associated with chronicity of illness. Mixed-handed patients were less likely to have a family history of psychotic illness than patients with strong right- or left-handedness. The results indicate a neurodevelopmental rather than a genetic origin for anomalous lateralization in schizophrenia.

Keywords: Ambidexterity; Laterality; Asymmetry; Chronicity; Nonfamilial schizophrenia; Neurodevelopment

1. Introduction

Functional differences between the left and right hemispheres of the brain have been recognized ever since Broca's discovery that lesions producing aphasia were almost always located in the left hemisphere. It is known that the left hemisphere is relatively specialized for language while the right hemisphere is relatively specialized for visuospatial functions. Structural asymmetries are also present in the human brain; for example, the planum temporale, an area of the temporal lobe concerned with language, has been found to be larger on the left side in 65% of normal brains studied (Geschwind and Levitsky, 1968).

There is evidence from many sources of an anomaly in the lateralization process in schizophrenia (Crow, 1990). Reversal of normal structural asymmetries has been found in the brains of schizophrenic patients in the occipital (Luchins et al., 1982) and temporal (Crow et al., 1989) regions. Asymmetry of the sylvian fissure (Falkai et al., 1992), asymmetry of dopamine D_2 receptor density

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(Reynolds et al., 1987), and increased fluctuating asymmetry of dermatoglyphic traits (Markow and Wandler, 1986) have also been reported in schizophrenic patients.

Many explanations for the anomalous patterns of lateralization in schizophrenia have been proposed, although none have yet been validated. Crow et al. (1989) suggest that schizophrenia may be related to an anomaly of the *right-shift* gene (or cerebral dominance gene), which is thought to control cerebral asymmetries and handedness (Annett, 1985) and which is theorized to shift the distribution of right-left differences about one standard deviation to the right. Bracha (1991) proposes that an environmental insult, particularly a second trimester prenatal injury, would be sufficient to cause the structural asymmetry seen in schizophrenia without invoking genetic factors. During fetal brain development, the left hemisphere lags behind the right, only catching up in the third trimester. Therefore, an insult that occurred during the second trimester would not affect both hemispheres equally, and the cerebral asymmetry seen in some schizophrenic patients may be a marker of early brain injury. Markow (1992) proposes a 'developmental instability' model: asymmetry or the lack of normal asymmetries is a measure of the developmental instability of an organism that is influenced by both genetic factors (homozygosity) and nongenetic factors such as exposure to environmental stressors during development or maternal stress during pregnancy.

Handedness is a subtle indicator of cerebral asymmetry, and in the absence of a direct test of cerebral dominance, it may be used to test some of the predictions from these models. Up to now, most studies have viewed handedness as a dichotomous variable in which an individual is either right- or left-handed. However, the current concept is that handedness can be viewed as a continuum from strong right-handedness to strong left-handedness with varying degrees of ambidexterity in between these extremes (Coren and Halpern, 1991). Deviations from the usual trajectory of lateralization would cause a shift along the continuum away from strong dextrality (for 80–90% of the population) rather than an abrupt change to left-handedness. In those for whom left-handedness is the normal preference pattern, a disruption in the lateralization process would cause a shift toward dextrality; because of the smaller numbers involved, however (approximately 10% of the population), such a shift would be more difficult to detect at the population level. The overall effect of a pathological process affecting lateralization in a certain group of individuals would therefore be a shift away from dextrality. Most studies of handedness and schizophrenia, using a dichotomous classification, have found that schizophrenic patients have an increased rate (~20%) of nonright-handedness compared with normal subjects (Dvirskii, 1976; Gur, 1977; Chagule and Master, 1981; Nasrallah et al., 1981, 1982), with two exceptions (Fleminger et al., 1977; Taylor et al., 1980), which found an excess of dextrality. Using an intermediate category of mixed-handedness, Green et al. (1989) found an increase of mixed-handedness in schizophrenia with no increase in pure left-handedness. Mixed-handedness has also been found to be more common in schizotypal personality disorder (Kim et al., 1992) and in psychometrically defined psychosis-prone individuals (Chapman and Chapman, 1987) than in the general population.

Twin studies of handedness and psychosis have yielded inconsistent results: Boklage (1977) reported that if both twins were dextral and one schizophrenic, the probability of concordance for schizophrenia was approximately 90%. If the twins were discordant for handedness, however, the concordance for schizophrenia dropped to 25%, and the sinistral twin was likely to be schizophrenic, with a milder illness than that occurring in dextral monozygotic twins. Boklage concluded that abnormalities of embryonic symmetry development are reflected simultaneously in the twinning process itself — in the abnormal laterality found in all monozygotic twins (incidence of sinistrality 20–35%) and in the etiology of schizophrenia in this population. This hypothesis received support from Luchins et al. (1980), who also observed that discordance for psychosis in monozygotic twins was related to discordance for right-handedness, and that nonright-handedness was associated with a milder form of schizophrenia, but Lewis et al.

(1989) and Torrey et al. (1993) failed to replicate these findings in subsequent studies of psychotic monozygotic twin pairs.

In the study outlined below, we set out to test the hypothesis that schizophrenic patients would have higher rates of mixed-handedness than normal comparison subjects, but would not have higher rates of pure left-handedness. To investigate the relationship between mixed-handedness and genetic vulnerability, we hypothesised that mixed-handed schizophrenic patients would be more likely to have a family history of schizophrenia than lateralized patients (on the assumption that a genetic process was the principal causative agent for the laterality deficit). Our third aim was to investigate whether mixed-handedness was associated with a milder or more severe form of schizophrenia as indicated by age at onset and chronicity.

2. Methods

2.1. Subjects

Subjects comprised 96 schizophrenic patients (69 males and 27 females) who met *DSM-III* criteria for schizophrenia (American Psychiatric Association, 1980) and 43 normal subjects (21 males and 22 females). Patients were drawn from two centers: the inpatient and long-stay hostel units of a large rural psychiatric hospital ($n = 51$) and an outpatient clinic at an urban community psychiatric center ($n = 45$). Normal subjects were recruited from medical, secretarial, and domestic staff members in both centers, and none of them had a personal or family history of psychiatric illness. The schizophrenic group was older than the control group: the mean age for the patients was 44.3 years (SD = 15.5), while that for the normal subjects was 33.4 years (SD = 11.1).

2.2. Case note review

Case note reviews for the schizophrenic group were carried out by two raters, without knowledge of the assessment of handedness. Information was extracted from the notes on age of onset (defined as age at first contact with psychiatric services), number and duration of hospital admissions, and family psychiatric history. Chronicity was defined as more than 1 year of continuous hospitalization. Information from the case notes about family psychiatric history was supplemented by inquiries to community psychiatric nurses and consultant psychiatrists in the area. Family history of psychiatric illness was divided into two categories: narrow band (first-degree relative with schizophrenia) and broad band (first- or second-degree relative with affective psychosis or second-degree relative with schizophrenia).

2.3. Handedness assessment

Handedness was assessed with the Edinburgh Handedness Inventory (EHI; Oldfield, 1971), a well-validated and reliable instrument for measuring handedness (Bryden, 1977) that has been standardized on a young adult population. The EHI uses a numerical scoring system unlike the classification system used for the Annett questionnaire (Annett, 1970) and is therefore more suitable for establishing where an individual lies on the continuum of handedness. The EHI gives a somewhat greater dispersion of subjects, with fewer congregated at the extremes of left and right than does the Annett questionnaire. Retest reliabilities are similar for the two questionnaires (0.80–0.86).

All patients were asked to demonstrate how they would perform each of the following 10 tasks: writing, drawing, throwing, using scissors, holding a toothbrush, using a knife, using a spoon, holding a broom (upper hand), striking a match, and opening a box (lid). The hand used was noted, and the subject was asked whether preference for right or left hand was strong (2 points) or weak (1 point) for each task. This method of assessing handedness therefore combined demonstration techniques (for assessment of direction of handedness) and self-report (for assessment of strength and consistency of response). The assessment was performed only once for each subject. Handedness was scored numerically for each subject according to the method proposed by Oldfield (1971): that is, subtract the score for the left hand from the score for the right hand, divide by the sum of both, and multiply by 100. This score, called the laterality quotient (LQ), can range along a continuum from +100 (totally right-handed) to

−100 (totally left-handed) with varying degrees of mixed-handedness between the two extremes.

2.4. Analysis

Subjects were divided into right-, mixed-, and left-handed groups on the basis of LQ scores, according to a 100% criterion previously used in schizophrenic populations (Green et al., 1989); that is, subjects scoring +100 were classified as right-handed, while subjects scoring −100 were classified as left-handed. Subjects with LQ scores between these two points were classified as mixed-handed. Results were analyzed with Mantel-Haenszel odds ratios, adjusted for age and sex, and 2 × 2 χ^2 analysis, with two-tailed probability tests, with the statistical software Statistical Package for the Social Sciences for Windows and STATA.

3. Results

Table 2 presents the results of the handedness assessment. Eighty-one percent of normal subjects, and 58% of patients use the right hand for all tasks. Seven percent of normal subjects, and 5% of patients are exclusive left-handers. The most striking difference between the groups is the high rate of mixed-handedness in the schizophrenic group compared with the normal group (36.5% vs. 11.6%). To test the hypothesis that schizophrenic patients would show a deficit of lateralization in both directions, the right- and left-handers were combined to form a 'lateralized' group. The crude Mantel-Haenszel odds ratio for mixed-handedness in the patients compared with the normal subjects is 4.4 (95% confidence interval [CI] = 1.5–13.9; χ^2 = 8.9, P = 0.003). The Mantel-Haenszel odds ratio remained significant after adjustment for the confounding effects of sex and age (3.06, 95% CI = 1.06–8.8; χ^2 = 4.8, P = 0.03).

Table 1
Handedness categories in schizophrenic and normal subjects (by sex)

Handedness	Schizophrenic patients		Normal subjects	
	n (%)	% Male	n (%)	% Male
Right	56 (58.3)	66	35 (81.4)	46
Mixed	35 (36.4)	83	5 (11.6)	80
Left	2 (5.2)	60	3 (7.0)	33
Total	96	72	43	49

Table 2
Odds ratios (OR) for mixed-handedness in schizophrenic patients compared with normal subjects (crude OR and OR adjusted for potential confounding factors)

Adjusted for	OR	95% confidence interval	χ^2	P (2-tailed)
Crude OR	4.36	1.50–13.93	8.8	0.003
Sex	3.65	1.25–10.64	6.5	0.01
Age	3.94	1.25–12.41	6.4	0.01
Sex + age	3.06	1.06–8.79	4.8	0.03

To test the hypothesis that mixed-handed patients are more likely to have a family history of schizophrenia, the odds ratios for mixed-handedness in patients with a family history of psychosis (narrow band only and narrow + broad bands) compared with patients with no known family history were calculated (see Table 2). Contrary to our hypothesis, mixed-handed patients were less likely to have a family history of psychosis than were strongly lateralized patients. This effect was the same when first-degree relatives only were examined (odds ratio = 0.41, 95% CI = 0.13–1.32; P = 0.21) and when both first- and second-degree relatives were included (odds ratio = 0.44, 95% CI = 0.17–1.08; P = 0.06). This finding was not statistically significant but approached statistical significance when both first- and second-degree relatives were included due to the increased numbers in the cells. There was a trend (odds ratio = 2.5, 95% CI = 0.8–8.6; P = 0.07) for mixed-handed patients to be males.

To investigate the third hypothesis, that mixed-handed patients would have a more severe clinical condition, we examined two markers of disease severity: mean age at onset of schizophrenia and chronicity. The mean age at onset for mixed-handed patients was 24.6 years (SD = 6.46), and the mean age at onset for the lateralized patients was 25.2 (SD = 7.63). Comparison of these means by t test did not show a significant difference

Table 3
Family history, sex, and chronicity in mixed-handed patients compared with lateralized patients

	Mixed	Lateralized (R+L)	Odds ratio	95% confidence interval	χ^2	*P* value
Family history of psychosis[a]						
None	19	21				
Narrow band only	6	16	0.41	0.13–1.32	2.41	0.12
None	19	21				
Narrow band + broad band	13	33	0.44	0.17–1.08	3.39	0.06
Sex						
Male	29	40				
Female	6	21	2.54	0.83–8.6	3.29	0.07
Chronicity						
No	9	36				
Yes	26	25	4.16	1.57–10.9	9.8	0.002

[a]Reliable information on family psychiatric history not available for 10 patients.

(t = 0.34, P = 0.74). Chronicity of illness was strongly associated with mixed handedness. Chronic, institutionalized patients were four times more likely than noninstitutionalized patients to be mixed-handed (odds ratio = 4.16, 95% CI = 1.5–10.9; P = 0.002; see Table 3). This effect remained significant after adjustment for age and sex (adjusted odds ratio = 5.1, 95% CI = 1.5–11.1; P = 0.003). When patterns of handedness were examined for outpatients only (i.e., good prognosis patients) compared with normal subjects, no significant excess of mixed-handedness was found (9/45 vs. 5/45; χ^2 = 1.35, P = 0.24).

4. Discussion

The schizophrenic patients were three times more likely to be mixed-handed than were members of the comparison group. Thus, the present study provides support for previous findings (Walker and Birch, 1970; Wahl, 1976; Nasrallah et al., 1981; Green et al., 1989) of a link between schizophrenia and mixed-handedness. We did not find an increase in pure left-handedness among our schizophrenic group. The difference in levels of mixed-handedness between schizophrenic and normal subjects found in this study (36.4% vs. 11.6%) is similar to the findings of Green et al. (1989) and Nelson et al. (1993), who observed that 38.7% and 43% of schizophrenic patients, respectively, were mixed-handed compared with 14.3% of normal subjects. Both studies also used demonstration of hand preference in the assessment of handedness.

The increase in mixed-handedness among the schizophrenic group supports the theory that there is a defect in the normal process of lateralization. If this were purely a genetic process, as suggested by Crow et al. (1989), the mixed-handed schizophrenic patients would be expected to be more likely to have a family history of schizophrenia. Our data did not support such an association; in fact, mixed-handed patients were only half as likely to have a positive family history of psychiatric illness as the lateralized patients (odds ratio = 0.43, 95% CI = 0.17–1.08). Although data on family history of psychiatric illness were not available for 10 patients, our method of collecting family history (case note review) is more likely to bias toward reporting patients with a positive family history and may therefore tend to underestimate the negative association between mixed-handedness and family history of schizophrenia. Shimizu et al. (1985) also found that nonright-handed schizophrenic patients were less likely to have a positive family history of schizophrenia than right-handed patients. This association suggests that some factor other than a purely genetic factor may be involved in the etiology of schizophrenia in mixed-handed individuals.

There was a strong association between mixed-handedness and chronicity, which remained after adjustment for potential confounding effects of age and sex. In fact, the excess of mixed-handedness in our sample appears to be confined to the chronic institutionalized patients. This is counter to the findings of Boklage (1977) and Luchins et al. (1982), who reported that nonright-handedness was associated with a milder form of schizophrenia, but it is in agreement with the findings of Andreasen and Olsen (1983), who reported that sinistrality in schizophrenic patients was associated with chronicity and negative symptomatology. Because of the high proportion of chronic patients in our sample (51%), there may have been a bias toward detecting higher rates of mixed-handedness than would have been found in a more representative catchment-area sample. Varying proportions of chronic patients in the samples of previous studies of handedness in schizophrenia could have accounted for some of the variability in results.

It is unlikely that the process of institutionalization itself would cause a change in handedness patterns as it has been shown that hand preferences remain stable from the early school years (Annett, 1970; Kilshaw and Annett, 1983). It is more likely that the shift in handedness in this group may be a marker for some form of neurological insult that has led to worse prognosis in the schizophrenic group. There is now a large body of literature showing a relationship between birth stress and deviations from the population norm of right-handedness (Coren and Halpern, 1991). A study of children with extremely low birth weights found that 54% were nonright-handed compared with 8% among those with higher birth weights (O'Callaghan et al., 1987). Bakan et al. (1973) suggested that the actual causal factor is oxygen deficiency induced by perinatal stress, and that nonright-handedness may be a marker for mild or otherwise difficult-to-detect instances of neuropathology. Schizophrenic patients are more likely to have a history of birth complications than are other psychiatric patients, siblings, or normal comparison subjects (Murray et al., 1988). Anomalous handedness may therefore be a marker of neural pathology and a 'neurodevelopmental' type of schizophrenia with a poor prognosis. The clinical profile of the mixed-handed patients in our study has many features in common with the characteristics of the 'neurodevelopmental' type of schizophrenia, as proposed by Murray et al. (1992): poor clinical course, lack of family history of schizophrenia, and male sex. A second-trimester insult such as maternal influenza (O'Callaghan et al., 1991), which has been associated with increased risk of schizophrenia in adulthood, could potentially interrupt the process of cerebral lateralization and give rise to mixed-handedness. A neurodevelopmental basis for mixed-handedness is supported by the excess of mixed-handedness in autism (Sank and Sank, 1979) and dyslexia (Hynd and Semrud-Clikeman, 1989).

Future studies in the area of lateralization and schizophrenia should continue to examine mixed-handedness separately. Detailed information on family psychiatric history and the inclusion of measures of cognitive ability and neuroimaging could help to answer some of the questions raised by these results.

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